

RIO DE JANEIRO ALFONSOS, Brazil

WMO: 837480

Lat: 22.875S

Lon: 43.385W

Elev: 34

StdP: 100.92

Time Zone: -3.00 (SAB)

Period: 98-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	13.9	9.9	7.6	19.9	11.0	8.2	19.9	7.7	25.0	6.6	24.5	0.9	240	0.282

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.1	37.2	24.9	36.1	24.7	35.0	24.7	27.5	32.9	26.9	32.3	26.2	31.4	3.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.6	30.6	25.2	20.5	29.4	25.0	20.1	29.1	87.4	33.0	84.5	32.5	81.6	31.2	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.5	5.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.5	27.9	28.6	27.1	24.6	22.2	21.4	21.2	22.0	23.3	24.4	25.0	26.7	(6)
(7)		DBStd	3.50	2.61	2.21	2.57	2.30	2.21	2.26	2.30	2.37	2.74	2.54	2.93	2.72	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	12	0	0	0	0	1	3	4	2	1	0	0	0	(9)
(10)		CDD10.0	5298	554	521	530	438	378	341	348	373	400	447	452	518	(10)
(11)		CDD18.3	2269	296	287	272	188	121	94	93	117	151	189	202	259	(11)
(12)		CDH23.3	20780	3273	3442	2717	1426	814	593	676	931	1171	1478	1684	2574	(12)
(13)		CDH26.7	8954	1614	1783	1182	485	247	174	219	348	488	588	684	1140	(13)
(14)	Wind	WSAvg	2.1	2.2	2.2	2.0	1.9	1.7	1.6	1.7	2.0	2.3	2.5	2.5	2.3	(14)
(15)	Precipitation	PrecAvg	1340	216	136	164	98	68	47	40	37	78	96	157	202	(15)
(16)		PrecMax	1873	378	312	374	264	133	200	114	134	151	250	333	478	(16)
(17)		PrecMin	580	61	7	36	21	27	2	5	9	23	22	42	73	(17)
(18)		PrecStd	297	85	76	92	55	31	36	27	28	37	59	70	85	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	39.0	37.2	35.0	33.8	32.8	33.0	34.2	36.9	36.8	37.2	37.8	(19)
(20)			MCWB	24.8	25.2	26.1	23.8	23.3	21.1	21.3	21.0	21.5	23.1	24.1	24.8	(20)
(21)		2%	DB	36.8	37.1	35.8	32.8	31.2	30.8	31.0	32.2	34.0	34.0	34.9	35.7	(21)
(22)			MCWB	25.3	25.0	25.7	23.9	22.1	21.2	20.5	20.9	21.9	22.9	24.1	25.1	(22)
(23)		5%	DB	35.2	35.9	34.0	31.1	29.2	28.9	29.2	30.4	31.8	31.9	32.9	34.0	(23)
(24)			MCWB	25.2	25.0	25.4	23.8	21.6	20.8	20.4	20.8	21.7	22.8	23.9	25.0	(24)
(25)		10%	DB	33.9	34.2	32.2	29.3	27.2	26.9	27.2	28.2	29.2	30.0	30.8	32.2	(25)
(26)			MCWB	25.0	24.9	25.1	23.3	20.8	20.6	20.5	20.6	21.5	22.6	23.7	24.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	28.5	28.4	26.8	25.1	23.7	24.2	24.5	24.9	25.3	26.1	27.6	(27)
(28)			MCDWB	34.1	34.5	34.1	30.9	31.2	28.7	29.2	30.4	31.3	31.3	32.7	33.2	(28)
(29)		2%	WB	26.8	27.6	27.3	25.5	23.6	22.5	22.5	23.2	23.6	24.4	25.3	26.6	(29)
(30)			MCDWB	33.4	32.8	32.4	29.4	28.1	27.4	27.0	28.4	29.7	30.3	31.4	32.1	(30)
(31)		5%	WB	26.1	26.9	26.5	24.6	22.4	21.7	21.6	22.1	22.8	23.7	24.7	25.9	(31)
(32)			MCDWB	32.1	32.1	31.2	28.8	27.2	26.1	26.0	27.0	28.3	29.5	30.3	31.2	(32)
(33)		10%	WB	25.6	26.1	25.9	23.9	21.6	21.1	21.0	21.1	22.2	23.1	24.0	25.3	(33)
(34)			MCDWB	31.2	31.0	30.2	28.3	26.3	25.0	25.2	26.1	27.2	28.3	29.0	30.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.3	11.1	9.3	9.6	10.4	10.6	11.1	11.7	10.7	9.7	8.8	9.3	(35)
(36)			MCDBR	13.0	13.2	11.9	11.6	13.8	14.4	15.3	16.0	15.8	13.7	12.7	12.4	(36)
(37)			MCWBR	4.1	3.9	3.9	4.6	5.6	6.2	6.6	6.2	5.1	4.9	4.1	4.2	(37)
(38)		5% WB	MCDWB	11.0	10.3	9.6	9.7	11.8	10.9	11.9	12.5	11.7	11.1	10.5	10.2	(38)
(39)			MCWBR	4.3	4.2	3.8	4.5	5.3	5.4	5.9	5.9	4.8	4.7	4.0	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.413	0.410	0.409	0.399	0.382	0.381	0.382	0.414	0.494	0.469	0.434	0.425	(40)	
(41)		taud	2.470	2.461	2.472	2.487	2.508	2.498	2.476	2.367	2.119	2.252	2.378	2.423	(41)	
(42)		Ebn at Noon	932	922	896	862	836	815	827	834	805	859	907	923	(42)	
(43)		Edh at Noon	119	118	112	102	92	89	94	113	157	144	130	125	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.01	6.25	5.34	4.68	3.72	3.47	3.60	4.27	4.59	5.15	5.30	5.87	(44)	
(45)		RadStd	0.88	0.64	0.45	0.34	0.20	0.26	0.31	0.33	0.59	0.61	0.44	0.53	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	N/A	N/A	-1.69	N/A	-1.40	-1.66	N/A	N/A	N/A	N/A

Nomenclature: See separate page